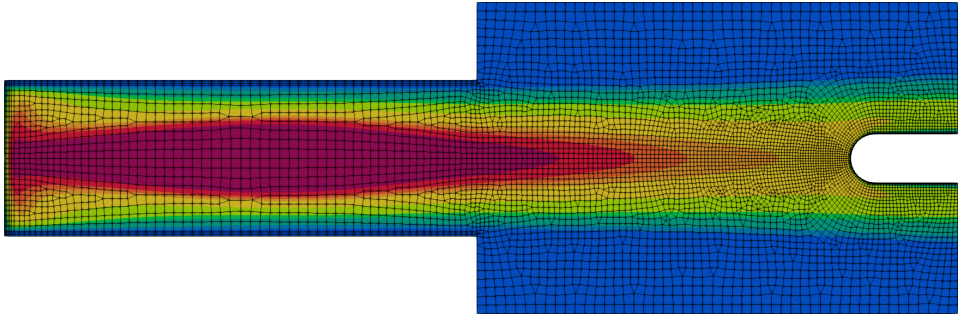


A Hybridized Discontinuous Galerkin Solver for Inductively Coupled Plasma

Nicolas Corthouts



Introduction

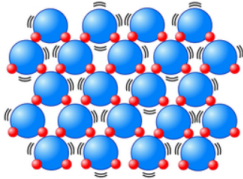
Au fait, qu'est-ce que j'ai fait pendant 4 ans?

J'ai développé un **solver numérique** pour les **plasmas à induction**.

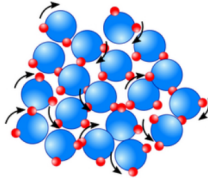


Etats de la matière: eau à pression atmosphérique

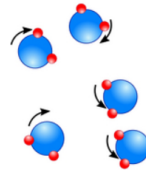
Solide



Liquide

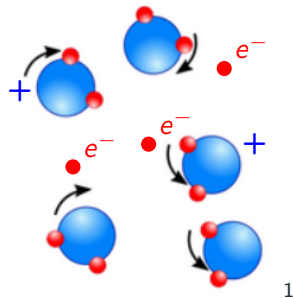
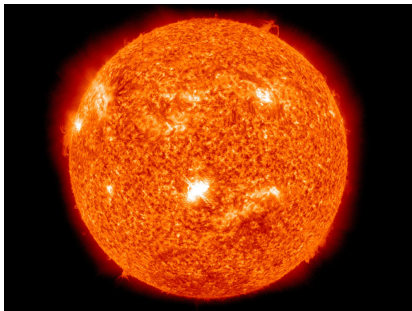


Gaz



Que se passe-t-il si on chauffe encore plus?

Plasma: le quatrième état de la matière



Les plasmas composent 90% de l'univers visible.

— Fusion nucléaire, médecine, métallurgie, lasers, création de microprocesseurs, ...

¹Courtesy of Wikipedia.

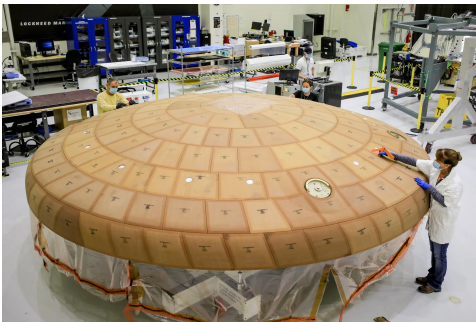
Plasma en réentrée atmosphérique



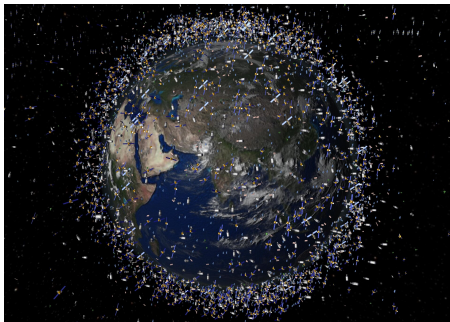
Vitesse de réentrée $\simeq 10 \text{ km s}^{-1}$: choc transformant l'air en plasma.

²Coutesy of NASA.

Système de protection thermique et destruction des déchets spatiaux



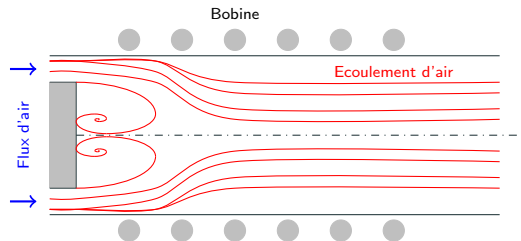
TPS



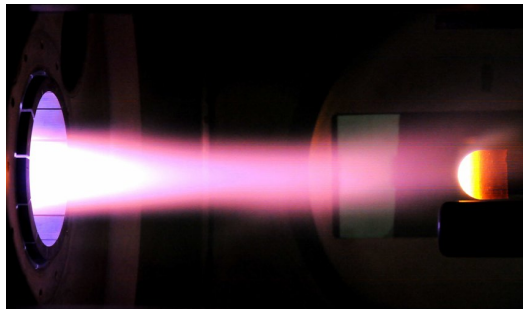
Déchets spatiaux

Expériences reproduisant les plasmas de réentrée atmosphérique.

Plasma à induction



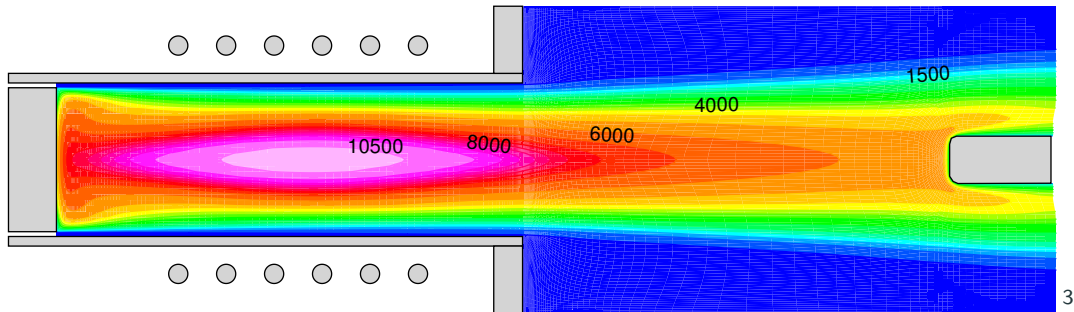
Torche



Chambre

Exemples: Plasmatron (VKI), Plasmatron X (Illinois), IPG (Russie), ...

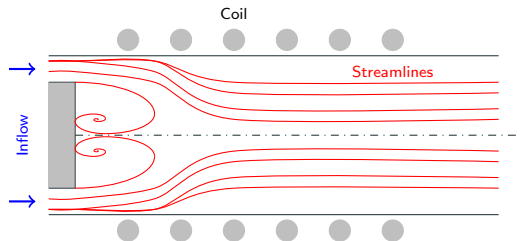
Plasma à induction: solveurs numériques



On propose un nouveau type de solveur numérique pour les plasmas à induction.

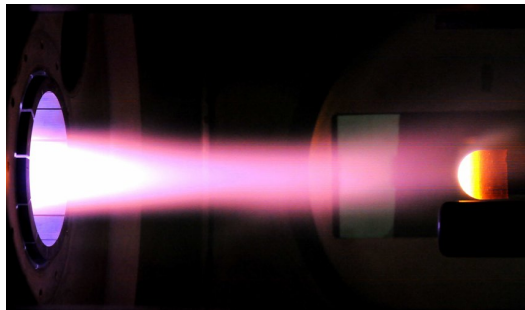
Thesis motivation

ICP facility: an overview



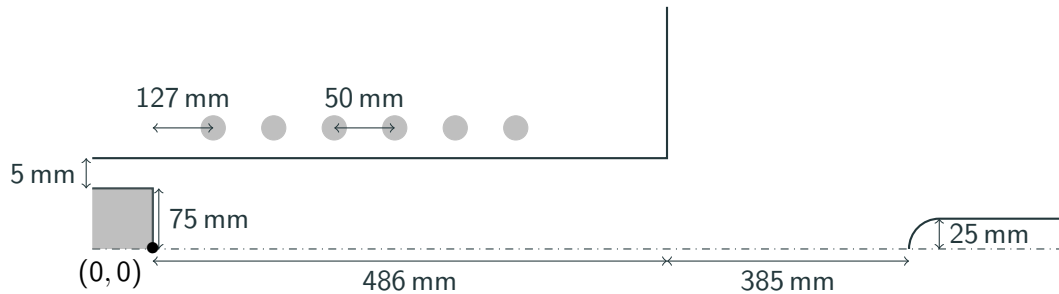
Torch

- $T \simeq 10\,000\text{ K}$, $p \in [1000\text{ Pa}; 100\,000\text{ Pa}]$.
- $Re \simeq 100$, $Ma \simeq 0.01$.
- Air, argon, carbon dioxide.



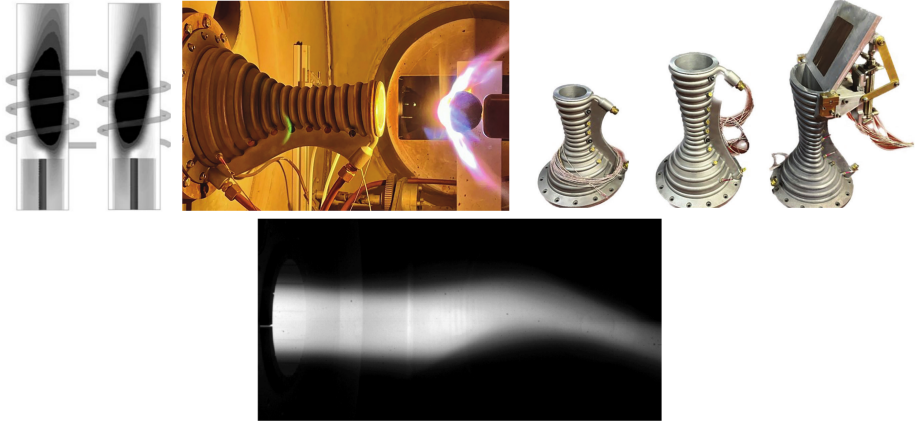
Chamber

- At LTE & non equilibrium.
- Local heat transfer principle.

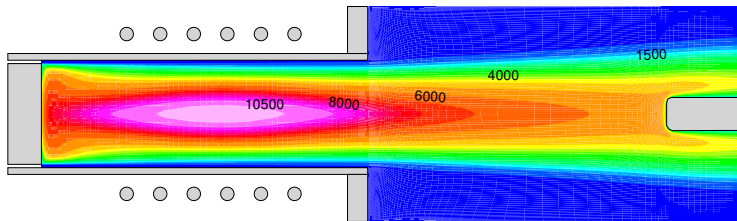


Large temperature gradients!

ICP simulations : complex physics at play

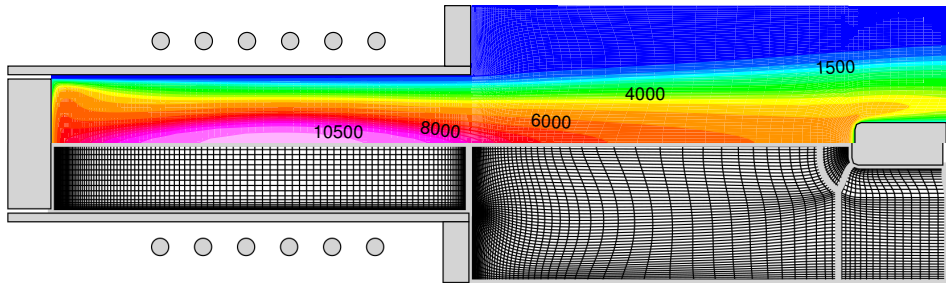


ICP : existing solvers



- Axisymmetric.
- Subsonic.
- Steady-state.
- Local thermodynamic equilibrium.
- No-radiation.
- Relatively slow convergence.
- Requires a constant attention during convergence processes.

Finite volume solver limitations.



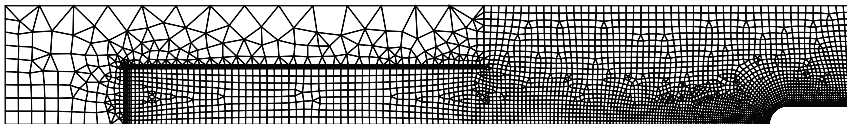
FV require a highly refined mesh in regions of large solution gradient.

FV can be very dissipative, "smoothing" the physics.

Solution procedure makes convergence slow and tedious...

Developing a hybridized discontinuous Galerkin solver of inductively coupled plasmas.

- Solution is not constant anymore on every element.
- Much greater mesh flexibility and (hopefully) less mesh elements.
- We hope to make a more robust solver.
- More precise (less dissipative) solver than finite volumes.



Research questions

With the objective of producing a new high order solver for inductively coupled plasma, we will try to answer the following questions:

- Q1 In addition of being precise, can a high-order solver be robust for inductively coupled plasma applications?**
- Q2 Is the developed solver user-friendly?**
- Q3 Can the new solver be easily adapted to the new experiments performed in ICP facilities?**

Model for inductively coupled plasma

Plasma governing equations : the Boltzmann equation

$$\mathcal{D}_i(f_{i,l}) = \partial_t f_{i,l} + \mathbf{c}_i \cdot \nabla f_{i,l} + \frac{\mathbf{F}_i}{m_i} \cdot \partial_{\mathbf{c}_i} f_{i,l} = \mathcal{S}_i + \mathcal{C}_i$$

$f_i(\mathbf{x}, \mathbf{c})$: probability density function.

$\mathcal{S}_i$: scattering collision operator.

$\mathcal{C}_i$: chemically reacting collision operator.

Provided $Kn \ll 1$ (continuous medium) + conservation of mass, momentum and energy, it is possible to simplify this system by integration over the velocity space.

Plasma collective behaviour

A plasma is a quasi-neutral gas composed of charged and neutral particles exhibiting a collective behaviour (through collisions = particles interaction).

Short ranged



Direct contact (local & binary)

Long ranged



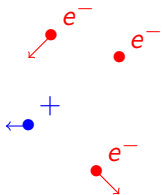
Electric force (distance and collective)

For scales $>$ Debye length ($1\text{ }\mu\text{m}$ in our case), $q \simeq 0$ due to electric force.

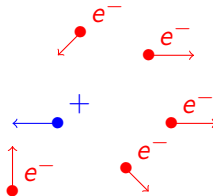
Cold plasmas and thermal equilibrium

"Cold plasmas" store energy in the free electrons. It is then transmitted during collisions to the ions and neutrals.

Inefficient collisions ($T_e \gg T_h$) Efficient collisions ($T_e \simeq T_h$)



Thermal non-equilibrium



Thermal equilibrium

Plasmas are in thermal equilibrium in this work.

Collisions may lead to chemical reactions.

If τ_{chem} and τ_{hydro} are characteristic chemical reaction and flow time respectively:

$\tau_{chem} \gg \tau_{hydro}$	$\tau_{chem} \simeq \tau_{hydro}$	$\tau_{chem} \ll \tau_{hydro}$
Chemical equilibrium	Chemical non-equilibrium	Frozen flow

Plasmas are in chemical equilibrium in this work.

**Thermal + Chemical equilibrium = Local Thermodynamic Equilibrium
(LTE)**

General equation for unmagnetized, optically thick plasma at LTE

$$\begin{cases} \partial_t \rho + \nabla \cdot (\rho \mathbf{v}) & = 0 \\ \partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v} + p \mathbb{I}) - \nabla \cdot \boldsymbol{\tau} & = \mathbf{j} \times \mathbf{B} \\ \partial_t \left(\frac{1}{2} \rho \|\mathbf{v}\|^2 + \rho e \right) + \nabla \cdot (\rho H \mathbf{v}) + \nabla \cdot (\mathbf{q} - \boldsymbol{\tau} \cdot \mathbf{v}) & = \mathbf{j} \cdot \mathbf{E} \end{cases}$$

Linked to Boltzmann equation

$$\rho = \sum_i \int m_i f_i d\mathbf{c}_i, \quad \rho \mathbf{v} = \sum_i \int m_i \mathbf{c}_i f_i d\mathbf{c}_i, \dots$$

General equation for unmagnetized, optically thick plasma at LTE

$$\begin{cases} \partial_t \rho + \nabla \cdot (\rho \mathbf{v}) & = 0 \\ \partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v} + p \mathbb{I}) - \nabla \cdot \boldsymbol{\tau} & = \mathbf{j} \times \mathbf{B} \\ \partial_t \left(\frac{1}{2} \rho \|\mathbf{v}\|^2 + \rho e \right) + \nabla \cdot (\rho H \mathbf{v}) + \nabla \cdot (\mathbf{q} - \boldsymbol{\tau} \cdot \mathbf{v}) & = \mathbf{j} \cdot \mathbf{E} \end{cases}$$

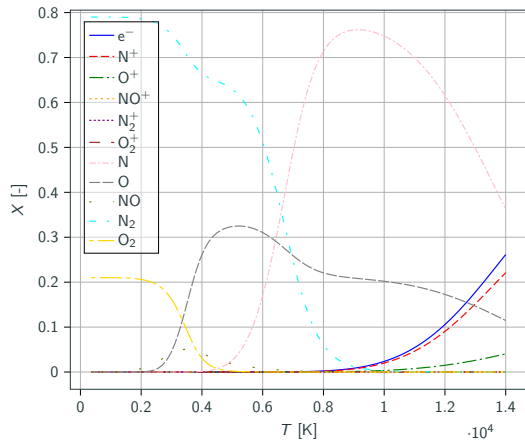
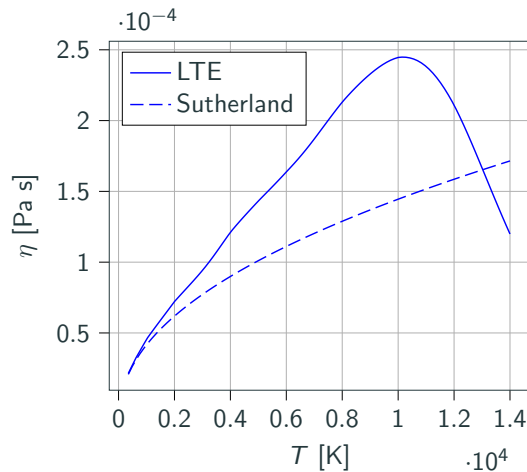
$$\boldsymbol{\tau} = \eta \left[\nabla \mathbf{v} + (\nabla \mathbf{v})^T - \frac{2}{3} (\nabla \cdot \mathbf{v}) \mathbb{I} \right]$$

$$\mathbf{q} = -\lambda \nabla T$$

$$\mathbf{j} = \sigma \mathbf{E} - \sum_{i,j \in S} n_i q_i D_{ij} \left[\frac{\nabla p_j}{n k_B T_h} - \frac{y_j p}{n k_B T_h} \nabla \ln(p) + \left(k_T^h + k_T^e \right) \nabla \ln(T) \right]$$

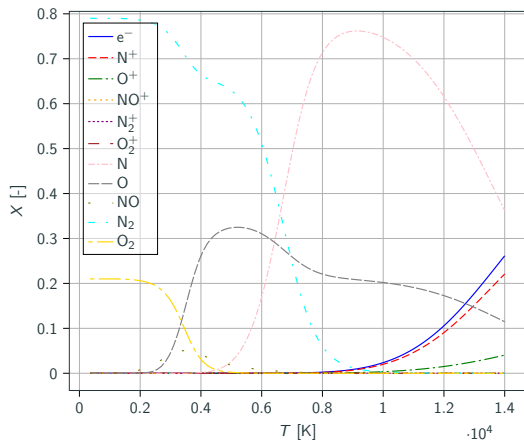
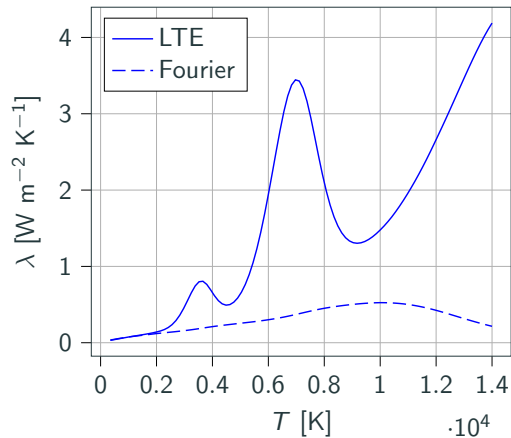
M⁺⁺utation
Multicomponent Thermodynamic And Transport properties for IONized gases in C++

Transport properties



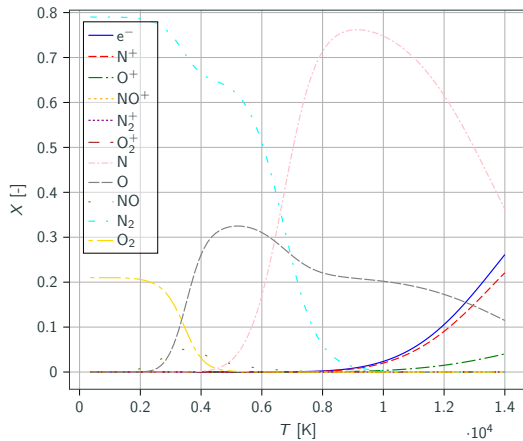
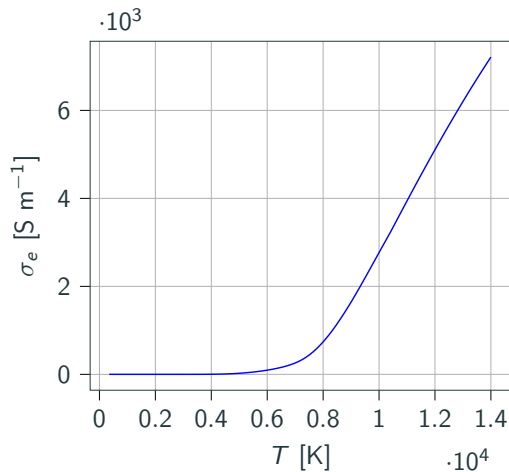
$$\eta \propto \sqrt{T}/\sigma$$

Transport properties



$$\lambda_r \propto \frac{dX}{dT}$$

Transport properties



$$\sigma_e \propto X_e$$

Navier-Stokes and Maxwell equations

$$\begin{cases} \partial_t \rho + \nabla \cdot (\rho \mathbf{v}) & = 0 \\ \partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v} + p \mathbb{I}) - \nabla \cdot \boldsymbol{\tau} & = \mathbf{j} \times \mathbf{B} \\ \partial_t \left(\frac{1}{2} \rho \|\mathbf{v}\|^2 + \rho e \right) + \nabla \cdot (\rho H \mathbf{v}) + \nabla \cdot (\mathbf{q} - \boldsymbol{\tau} \cdot \mathbf{v}) & = \mathbf{j} \cdot \mathbf{E} \end{cases}$$

$$\begin{aligned} \nabla \cdot \mathbf{E} &= 0 & \partial_t \mathbf{B} + \nabla \times \mathbf{E} &= \mathbf{0} \\ \nabla \cdot \mathbf{B} &= 0 & -\epsilon_0 \partial_t \mathbf{E} + \frac{1}{\mu_0} \nabla \times \mathbf{B} &= \mathbf{j} \end{aligned}$$

Hypothesis specific to ICP

- Quasi-steady & averaged over an oscillation period.

$$\partial_t f \int_0^{1/f} dt \simeq 0$$

- Axisymmetric

$$\partial_\theta \simeq 0$$

- No displacement current.

$$\epsilon_0 \partial_t \mathbf{E} \simeq 0$$

- Ambipolar electrostatic field ($j_z = j_r = 0$).

$$\mathbf{j} = \sigma_e \mathbf{E}_I$$

- Split of the induced electric field in a coil and plasma part

$$E_I = E_C + E_P$$

- The coils are thin parallel wires surrounding the facility.

$$\sum_{l=1}^{N_{coil}} \text{if } \mu_0 I_C \sqrt{\frac{r_0}{r}} \left[2 \frac{E_2(k_l)}{k_l} - E_1(k_l) \left(\frac{2}{k_l} - k_l \right) \right]$$

- We use phasor notation.

Equations governing the plasma

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{F}_L$$

$$\partial_t \left(\rho e + \frac{1}{2} \rho \|\mathbf{v}\|^2 \right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2} \rho \|\mathbf{v}\|^2 \mathbf{v} + p \mathbf{v} \right) = \nabla \cdot (\boldsymbol{\tau} \mathbf{v}) + \nabla \cdot \mathbf{q} + P_J$$

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$$\boldsymbol{\tau} = \eta \left(\nabla \mathbf{v} + \nabla \mathbf{v}^T \right) - \frac{2}{3} \eta \nabla \cdot \mathbf{v}$$

$$\mathbf{q} = -k \nabla T$$

$$F_z^L = \frac{\sigma_e}{4\pi f} \left[E_l^{lm} \partial_z E_l^{Re} - E_l^{Re} \partial_z E_l^{lm} \right]$$

$$F_r^L = \frac{\sigma_e}{4\pi f} \left[E_l^{lm} \frac{1}{r} \partial_r (r E_l^{Re}) - E_l^{Re} \frac{1}{r} \partial_r (r E_l^{lm}) \right]$$

$$P^J = \frac{\sigma_e}{2} \left[(E_l^{lm})^2 + (E_l^{Re})^2 \right]$$

Equations governing the plasma

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

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$$\partial_{zz}^2 E_P + \frac{1}{r} \partial_r (r \partial_r E_P) - \frac{E_P}{r^2} = i 2 \pi f \mu_0 \sigma_e (E_C + E_P)$$

μ_0 : magnetic permeability, σ_e : electron electric conductivity, f : induction frequency

Equations governing the plasma

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

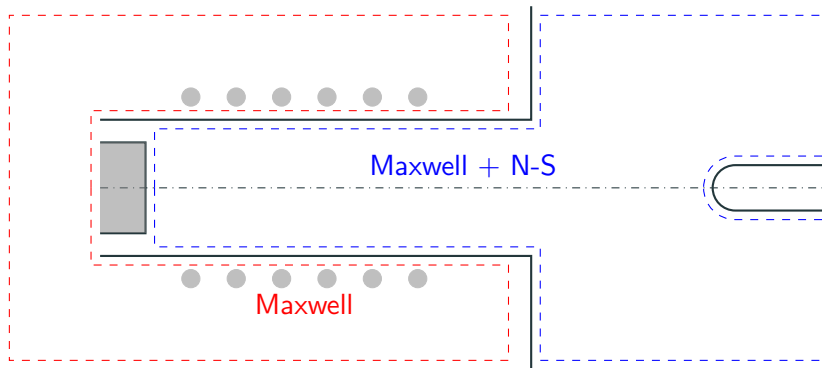
$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{F}_L$$

$$\partial_t \left(\rho e + \frac{1}{2} \rho \|\mathbf{v}\|^2 \right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2} \rho \|\mathbf{v}\|^2 \mathbf{v} + p \mathbf{v} \right) = \nabla \cdot (\boldsymbol{\tau} \mathbf{v}) + \nabla \cdot \mathbf{q} + P_J$$

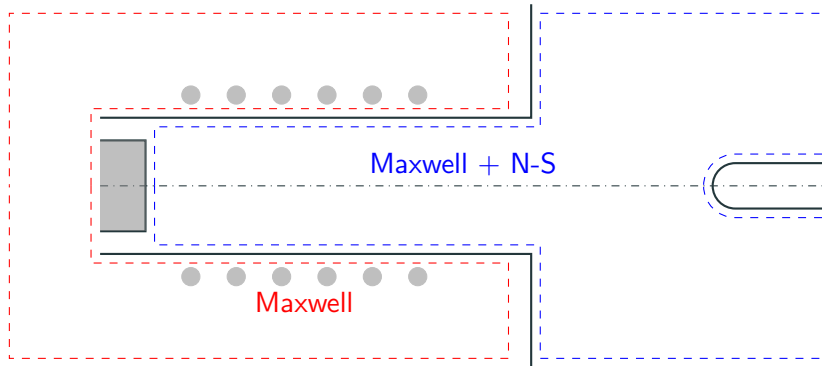
$$\partial_{zz}^2 E_P + \frac{1}{r} \partial_r (r \partial_r E_P) - \frac{E_P}{r^2} = i 2 \pi f \mu_0 \sigma_e (E_C + E_P)$$

μ_0 : magnetic permeability, σ_e : electron electric conductivity, f : induction frequency

Computational domain

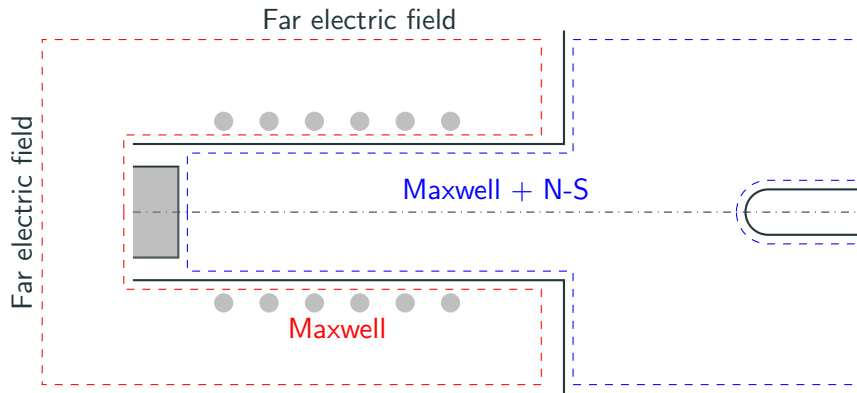


Computational domain



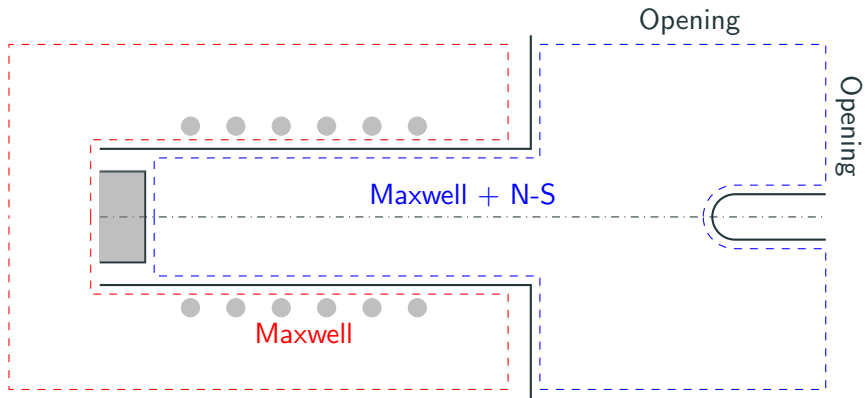
Symmetry axis $\partial_r p = 0$, $\partial_r u_z = 0$, $u_r = 0$, $\partial_r T = 0$ and $E_P = 0$.

Computational domain



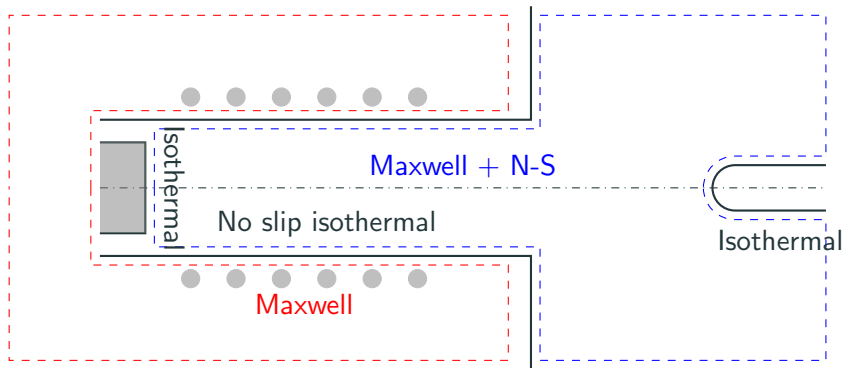
Vanishing far electric field $E_P = 0$.

Computational domain



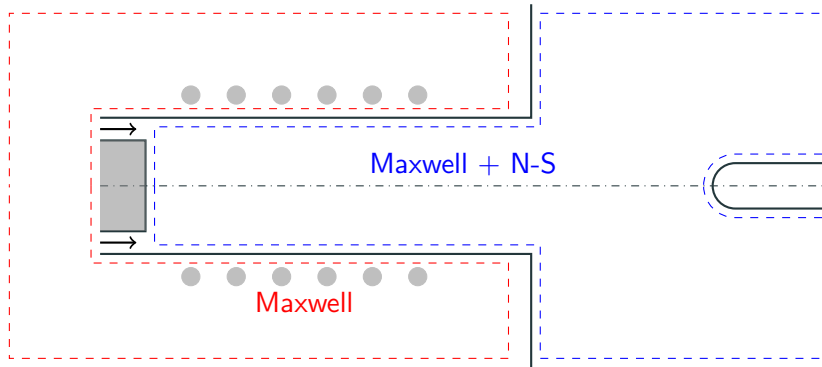
Openings $p = p_0$

Computational domain



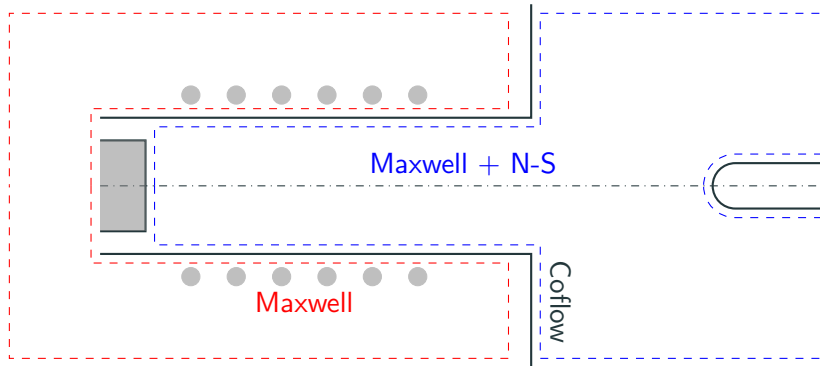
No-slip wall isothermal $T = T_{wall}$ and $\mathbf{u} = \mathbf{0}$.

Computational domain

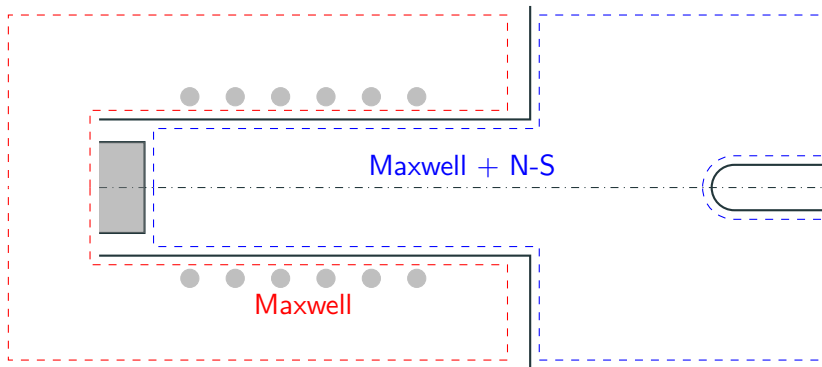


Inflow $u_z = U_{in}$, $u_r = 0$ and $T = T_{inlet}$.

Computational domain



Stabilizing co-flow $T = T_{wall}$, $\mathbf{u} = u_{coflow}\mathbf{e}_z$

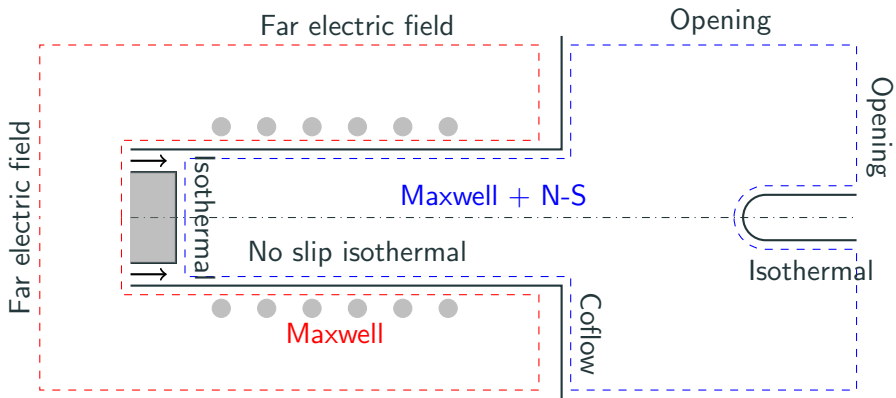


ICP and Maxwell interface

$$\nabla E_p^{Re,NS} \cdot \mathbf{n}^{NS} = \nabla E_p^{Re,MAX} \cdot \mathbf{n}^{MAX}$$

$$\nabla E_p^{Im,NS} \cdot \mathbf{n}^{NS} = \nabla E_p^{Im,MAX} \cdot \mathbf{n}^{MAX}$$

Computational domain



Hybridized discontinuous Galerkin method

Model problem

$$\begin{aligned}\partial_t \mathbf{w}(\mathbf{u}) + \nabla \cdot (\mathbf{F}_c(\mathbf{u}) - \mathbf{F}_v(\mathbf{u}, \nabla \mathbf{u})) &= \mathbf{S}(\mathbf{u}, \nabla \mathbf{u}) && \text{on } \Omega \\ \mathbf{u} &= \mathbf{u}_{bc} && \text{on } \partial\Omega_D \\ \mathbf{F}_v \cdot \mathbf{n} &= \mathbf{F}_{v,n,bc} && \text{on } \partial\Omega_N \\ \mathbf{u}(t=0) &= \mathbf{U} && \text{on } \Omega\end{aligned}$$

$\mathbf{F}_{c,v}$: convective and diffusive fluxes.

$\mathbf{S}$: source terms.

$\mathbf{w}$: conservative variables.

$\mathbf{u}, \mathbf{u}_{bc}$: solution and boundary condition.

Ω : domain of the problem.

$\partial\Omega_{D,N}$: domain boundary for Dirichlet and Neumann BC.

$\mathbf{U}$: initial data.

Model problem

$$\begin{aligned}\partial_t \mathbf{w}(\mathbf{u}) + \nabla \cdot (\mathbf{F}_c(\mathbf{u}) - \mathbf{F}_v(\mathbf{u}, \nabla \mathbf{u})) &= \mathbf{S}(\mathbf{u}, \nabla \mathbf{u}) && \text{on } \Omega \\ \mathbf{u} &= \mathbf{u}_{bc} && \text{on } \partial\Omega_D \\ \mathbf{F}_v \cdot \mathbf{n} &= \mathbf{F}_{v,n,bc} && \text{on } \partial\Omega_N \\ \mathbf{u}(t=0) &= \mathbf{U} && \text{on } \Omega\end{aligned}$$

Before going further, some notation:

$$\begin{aligned}(a, b)_K &= \int_K ab \, dV, \\ \langle a, b \rangle_{\partial K} &= \int_{\partial K} ab \, dS.\end{aligned}$$

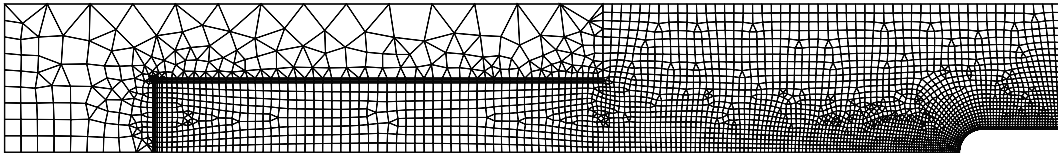
$$(\partial_t \mathbf{w} - \mathbf{S}, w)_\Omega - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_\Omega + \langle (\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}, w \rangle_{\partial\Omega} = 0, \quad \forall w \in L_2(\Omega)$$

Multiplying by a function $w \in L_2(\Omega)$ and integrating over the domain gives the **weak form**.

From weak form to HDG

$$(\partial_t \mathbf{w} - \mathbf{S}, w)_{\mathcal{T}} - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_{\mathcal{T}} + \sum_{K \in \mathcal{T}} \langle (\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}, w \rangle_{\partial K} = 0, \quad \forall w \in L_2(\Omega)$$

The domain Ω is split in a collection of elements $K \in \mathcal{T}$.



$$(\partial_t \mathbf{w} - \mathbf{S}, w)_{\mathcal{T}} - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_{\mathcal{T}} + \sum_{K \in \mathcal{T}} \langle (\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}, w \rangle_{\partial K} = 0, \quad \forall w \in W_h$$

We restrict w to a **finite dimensional** subset of $L_2(\Omega)$.

$$W_h = \{w \in L^2(\Omega) : w|_K \in \mathcal{P}^p(K), \forall K \in \mathcal{T}\} \subset L_2(\Omega)$$

From weak form to HDG

$$(\partial_t \mathbf{w} - \mathbf{S}, w)_{\mathcal{T}} - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_{\mathcal{T}} + \sum_{K \in \mathcal{T}} \langle \mathcal{H}(\mathbf{u}, \nabla \mathbf{u}), w \rangle_{\partial K} = 0, \quad \forall w \in W_h$$

For stabilizing the method, the convective and diffusive fluxes on the interface are approximated

$$(\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}_K \simeq \mathcal{H}$$

From weak form to HDG

$$(\partial_t \mathbf{w}_h - \mathbf{S}_h, w)_{\mathcal{T}} - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla w)_{\mathcal{T}} + \sum_{K \in \mathcal{T}} \langle \mathcal{H}(\mathbf{u}, \nabla \mathbf{u}), w \rangle_{\partial K} = 0, \quad \forall w \in W_h$$

The solution is also approximated by

$$\mathbf{u} \simeq \mathbf{u}_h \in W_h$$

From weak form to HDG

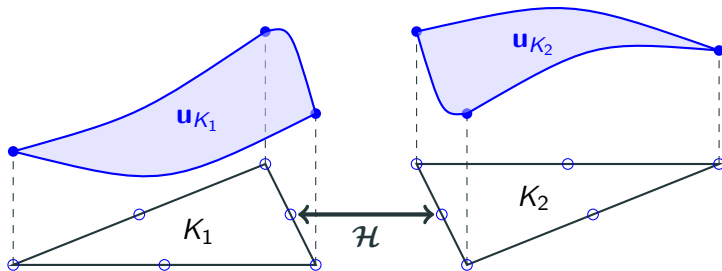
$$\begin{aligned} & (\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \mathcal{H}(\mathbf{u}_h, \nabla \mathbf{u}_h), \varphi_{K,i} \rangle_{\partial K} = 0, \\ & \forall \varphi_{K,i} \in W_h \end{aligned}$$

If the $\varphi_{K,i}$ is a local basis of W_h on K , $i \in \{1, 2, \dots, p\}$,

$$\mathbf{u} \simeq \mathbf{u}_h = \sum_{K \in \mathcal{T}} \sum_{i=1}^p \mathbf{u}_{K,i} \varphi_{K,i}$$

From weak form to HDG

$$(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \mathcal{H}(\mathbf{u}_h, \nabla \mathbf{u}_h), \varphi_{K,i} \rangle_{\partial K} = 0,$$
$$\forall \varphi_{K,i} \in W_h$$



From weak form to HDG

$$\begin{aligned}(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \mathcal{H}(\mathbf{u}_h, \mathbf{q}_h), \varphi_{K,i} \rangle_{\partial K} &= 0, \\ (\mathbf{q}_h, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_h, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \mathbf{u}_h, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} &= 0, \\ \forall \varphi_{K,i} \in W_h, \forall \boldsymbol{\tau}_{K,i} \in V_h\end{aligned}$$

With the solution gradient

$$\mathbf{q} = \nabla \mathbf{u} \simeq \mathbf{q}_h$$

with the subset

$$V_h = \left\{ \mathbf{v} \in L^2(\Omega) : \mathbf{v}|_K \in (\mathcal{P}^p(K))^D, \forall K \in \mathcal{T} \right\}$$

and local basis $\boldsymbol{\tau}_{K,i}$.

From weak form to HDG

$$\begin{aligned}(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \mathcal{H}(\boldsymbol{\lambda}_h, \mathbf{u}_h, \mathbf{q}_h), \varphi_{K,i} \rangle_{\partial K} &= 0, \\(\mathbf{q}_h, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_h, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_h, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} &= 0 \\ \forall \varphi_{K,i} \in W_h, \forall \boldsymbol{\tau}_{K,i} \in V_h\end{aligned}$$

$$\Gamma = \{e : e = K_i \cap K_j; \forall K_i, K_j \in \mathcal{T}, K_i \neq K_j\}.$$

$$M_h = \{\mu \in L^2(\Gamma) : \mu|_e \in \mathcal{P}^p(e), \forall e \in \Gamma\}$$

$$\boldsymbol{\lambda}_h = \sum_{e \in \Gamma} \sum_{i=1}^{N_p} \boldsymbol{\lambda}_{e,i} \mu_{e,i}$$

$$\begin{aligned}(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \mathcal{H}(\boldsymbol{\lambda}_h, \mathbf{u}_h, \mathbf{q}_h), \varphi_{K,i} \rangle_{\partial K} &= 0, \\(\mathbf{q}_h, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_h, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_h, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} &= 0, \\ \langle [[\mathcal{H}]], \mu_{e,i} \rangle_e &= 0, \\ \forall \mu_{e,i} \in M_h, \forall \varphi_{K,i} \in W_h, \forall \boldsymbol{\tau}_{K,i} \in V_h\end{aligned}$$

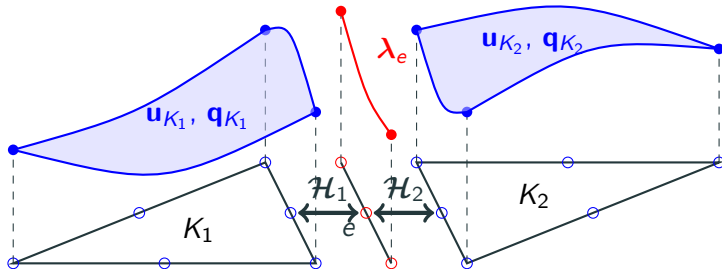
Equation for $\boldsymbol{\lambda}$: conservativity of the normal numerical flux across the interfaces.

The jump operator is given by

$$[[\mathbf{u}]] = \mathbf{u}_+ \cdot \mathbf{n}_+ + \mathbf{u}_- \cdot \mathbf{n}_-$$

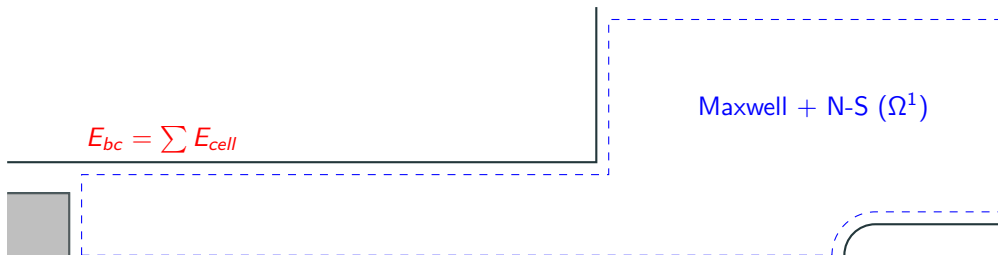
From weak form to HDG

$$\begin{aligned}
 &(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \mathcal{H}(\boldsymbol{\lambda}_h, \mathbf{u}_h, \mathbf{q}_h), \varphi_{K,i} \rangle_{\partial K} = 0, \\
 &(\mathbf{q}_h, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_h, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_h, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} = 0, \\
 &\langle [[\mathcal{H}]], \mu_{e,i} \rangle_e = 0, \\
 &\forall \mu_{e,i} \in M_h, \forall \varphi_{K,i} \in W_h, \forall \boldsymbol{\tau}_{K,i} \in V_h
 \end{aligned}$$



ICP as a multi-domain problem

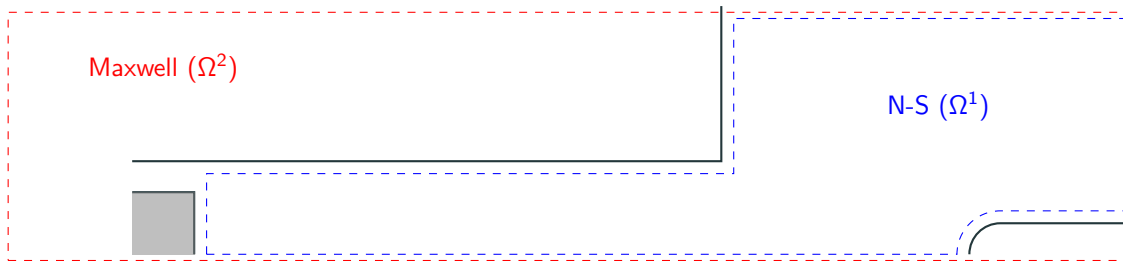
Single domain + integral boundary value



Compact but fills the jacobian matrix, making it incompatible with HDG.

ICP as a multi-domain problem

Two overlapping domain



Most widely used. Solve N-S while Maxwell frozen, and *vice versa*. Slow convergence.

ICP as a multi-domain problem

Multi-domain



Solves the system in a fully coupled manner, with two domains. The method we employ here.

Modification of the model problem

$$\partial_t \mathbf{w}^l(\mathbf{u}) + \nabla \cdot \left(\mathbf{F}_c^l(\mathbf{u}) - \mathbf{F}_v^l(\mathbf{u}, \nabla \mathbf{u}) \right) = \mathbf{S}^l(\mathbf{u}, \nabla \mathbf{u}), \quad \text{on } \Omega^l$$

$$\mathbf{u}^l = \mathbf{u}_{bc}^l, \quad \mathbf{x} \in \partial\Omega_d^l$$

$$\mathbf{F}_v^l \cdot \mathbf{n} = \mathbf{F}_{v,n,bc}^l, \quad \text{on } \partial\Omega_n^l$$

$$\mathcal{F}^{l,l'}(\mathbf{u}^l, \nabla \mathbf{u}^l, \mathbf{u}^{l'}, \nabla \mathbf{u}^{l'}) = 0, \quad \text{on } \gamma^{l,l'}, \quad l \neq l'$$

$$\mathbf{u}^l(t=0) = \mathbf{U}^l, \quad \text{on } \Omega^l$$

Modification of the model problem

$$\partial_t \mathbf{w}^l(\mathbf{u}) + \nabla \cdot \left(\mathbf{F}_c^l(\mathbf{u}) - \mathbf{F}_v^l(\mathbf{u}, \nabla \mathbf{u}) \right) = \mathbf{S}^l(\mathbf{u}, \nabla \mathbf{u}), \quad \text{on } \Omega^l$$

$$\mathbf{u}^l = \mathbf{u}_{bc}^l, \quad \mathbf{x} \in \partial\Omega_d^l$$

$$\mathbf{F}_v^l \cdot \mathbf{n} = \mathbf{F}_{v,n,bc}^l, \quad \text{on } \partial\Omega_n^l$$

$$\mathcal{F}^{l,l'}(\mathbf{u}^l, \nabla \mathbf{u}^l, \mathbf{u}^{l'}, \nabla \mathbf{u}^{l'}) = 0, \quad \text{on } \gamma^{l,l'}, \quad l \neq l'$$

$$\mathbf{u}^l(t=0) = \mathbf{U}^l, \quad \text{on } \Omega^l$$

Modification of the model problem

$$\begin{aligned}
 \partial_t \mathbf{w}^l(\mathbf{u}) + \nabla \cdot \left(\mathbf{F}_c^l(\mathbf{u}) - \mathbf{F}_v^l(\mathbf{u}, \nabla \mathbf{u}) \right) &= \mathbf{S}^l(\mathbf{u}, \nabla \mathbf{u}), & \text{on } \Omega^l \\
 \mathbf{u}^l &= \mathbf{u}_{bc}^l, & \mathbf{x} \in \partial\Omega_d^l \\
 \mathbf{F}_v^l \cdot \mathbf{n} &= \mathbf{F}_{v,n,bc}^l, & \text{on } \partial\Omega_n^l \\
 \mathcal{F}^{l,l'}(\mathbf{u}^l, \nabla \mathbf{u}^l, \mathbf{u}^{l'}, \nabla \mathbf{u}^{l'}) &= 0, & \text{on } \gamma^{l,l'}, l \neq l' \\
 \mathbf{u}^l(t=0) &= \mathbf{U}^l, & \text{on } \Omega^l
 \end{aligned}$$

For ICP,

$$\mathcal{F} = \begin{pmatrix} \nabla E_p^{Re,NS} \cdot \mathbf{n}^{NS} - \nabla E_p^{Re,MAX} \cdot \mathbf{n}^{MAX} \\ \nabla E_p^{Im,NS} \cdot \mathbf{n}^{NS} - \nabla E_p^{Im,MAX} \cdot \mathbf{n}^{MAX} \end{pmatrix}$$

Modification of the discrete formulation

$$\begin{aligned} & \left(\partial_t \mathbf{w}_h - \mathbf{S}_h^K, \varphi_{K,i} \right)_K - \left(\mathbf{F}_{h,c}^K - \mathbf{F}_{h,v}^K, \nabla \varphi_{K,i} \right)_K + \left\langle \mathcal{H}^K, \varphi_{K,i} \right\rangle_{\partial K} = 0 \\ & (\mathbf{q}_K, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_K, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_{\partial K}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} = 0 \\ & \langle [[\mathcal{H}]], \mu_{e,i} \rangle_{e \setminus \bar{\Gamma}} + \langle \tilde{\mathcal{F}}^e, \mu_{e,i} \rangle_{e \cap \bar{\Gamma}} = 0 \end{aligned}$$

with $\bar{\Gamma}$ the set of interfaces between subdomains.

Numerical fluxes : convection

Low mach requires to use preconditioned numerical flux

$$\mathcal{H} = \mathcal{H}_c - \mathcal{H}_v$$

$$\mathcal{H}_c = R\mathcal{H}_{c,n}(\mathbf{u}, \lambda) = R(\dot{m} + \dot{m}_p)\Psi_\lambda - R|\dot{m}|(\Psi_\lambda - \Psi_U) + \mathbf{P}$$

with

$$\Psi = \left(1 \quad v^\perp \quad v^\parallel \quad v_\theta \quad e + \frac{1}{2}||\mathbf{v}||^2 + \frac{p}{\rho} \right)^T$$

$$\mathbf{P} = \left(0 \quad p_\lambda \quad 0 \quad 0 \right)^T$$

$$\mathcal{H} = \mathcal{H}_c - \mathcal{H}_v$$

$$\mathcal{H}_v(\mathbf{u}, \lambda, \mathbf{q}) = \mathbf{F}_v(\lambda, \mathbf{q}) \cdot \mathbf{n} + \gamma(h)(\lambda - \mathbf{u})$$

with

$$\gamma = C \max_{f \in K} \left(\frac{1}{2\mathcal{V}} \sum_{f \in K} \mathcal{A} \right)$$

C depends on the diffusive transport coefficients.

Solution strategy

The HDG system is solved at steady-state such that

$$\begin{aligned}
 & \left(\cancel{\partial_t \mathbf{w}_h} - \mathbf{S}_h^K, \varphi_{K,i} \right)_K - \left(\mathbf{F}_{h,c}^K - \mathbf{F}_{h,v}^K, \nabla \varphi_{K,i} \right)_K + \left\langle \mathcal{H}^K, \varphi_{K,i} \right\rangle_{\partial K} = 0 \\
 & (\mathbf{q}_K, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_K, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_{\partial K}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} = 0 \\
 & \langle [[\mathcal{H}]], \mu_{e,i} \rangle_{e \setminus \bar{\Gamma}} + \langle \tilde{\mathcal{F}}^e, \mu_{e,i} \rangle_{e \cap \bar{\Gamma}} = 0
 \end{aligned}$$

If the system is written as $\mathcal{N} = 0$, then the Newton method solves

$$\mathcal{N}'(\mathbf{x}_k) \delta \mathbf{x}_k = -\mathcal{N}(\mathbf{x}_k)$$

with $\delta \mathbf{x}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$.

Solution strategy

The HDG system is solved at steady-state such that

$$\begin{aligned} \left(\frac{\delta \mathbf{w}_h^K}{\tau_{K,k}} - \mathbf{S}_h^K, \varphi_{K,i} \right)_K - \left(\mathbf{F}_{h,c}^K - \mathbf{F}_{h,v}^K, \nabla \varphi_{K,i} \right)_K + \left\langle \mathcal{H}^K, \varphi_{K,i} \right\rangle_{\partial K} &= 0 \\ (\mathbf{q}_K, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_K, \nabla \boldsymbol{\tau}_{K',i})_K + \langle \boldsymbol{\lambda}_{\partial K}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} &= 0 \\ \langle [[\mathcal{H}]], \mu_{e,i} \rangle_{e \setminus \bar{\Gamma}} + \langle \tilde{\mathcal{F}}^e, \mu_{e,i} \rangle_{e \cap \bar{\Gamma}} &= 0 \end{aligned}$$

If the system is written as $\tilde{\mathcal{N}} = 0$, then the Newton method solves

$$\tilde{\mathcal{N}}'(\mathbf{x}_k) \delta \mathbf{x} = -\tilde{\mathcal{N}}(\mathbf{x}_k)$$

with $\delta \mathbf{x} = \mathbf{x}_{k+1} - \mathbf{x}_k$.

In order to stabilize the solver, a fictive temporal term is added.

Damping term

$$\frac{\delta \mathbf{w}_h}{\tau_K} = \frac{\mathbf{w}(\mathbf{u}_{h,k+1}) - \mathbf{w}(\mathbf{u}_{h,k})}{\tau_{K,k}}$$

with

$$\tau_{K,k} = CFL_k \frac{h_K}{v_K} = CFL_k \frac{V_K}{\int_{\partial K} \lambda_{\max} dS}$$

and the following CFL evolution law

$$CFL_k = \min \left(CFL_0 \left(\frac{||\tilde{\mathcal{N}}||_2^0}{||\tilde{\mathcal{N}}||_2^k} \right)^\alpha, CFL_{\max} \right)$$

We only use convective eigenvalues here. The diffusive could be use to enhance the convergence.

Matrix form of the system

$$\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\tilde{\mathcal{N}}(\mathbf{x}_k)$$

Matrix form of the system

$$\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\tilde{\mathcal{N}}(\mathbf{x}_k)$$

$$\underbrace{\begin{pmatrix} A & B & R \\ C & D & S \\ L & M & N \end{pmatrix}}_{\tilde{\mathcal{N}}'} \overbrace{\begin{pmatrix} \delta Q \\ \delta U \\ \delta \Lambda \end{pmatrix}}^{\delta \mathbf{x}} = \underbrace{\begin{pmatrix} F \\ G \\ H \end{pmatrix}}_{-\tilde{\mathcal{N}}},$$

Matrix form of the system

$$\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\tilde{\mathcal{N}}(\mathbf{x}_k)$$

The system can be split into two parts

$$\underbrace{\begin{pmatrix} A & B \\ C & D \end{pmatrix}}_{\text{Block diagonal}} \begin{pmatrix} \delta Q \\ \delta U \end{pmatrix} = \begin{pmatrix} F \\ G \end{pmatrix} - \begin{pmatrix} R \\ S \end{pmatrix} \delta\Lambda$$

and

$$\begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} \delta Q \\ \delta U \end{pmatrix} + N\delta\Lambda = H$$

Matrix form of the system

$$\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\tilde{\mathcal{N}}(\mathbf{x}_k)$$

By eliminating δU and δQ , one gets the **global** system

$$\left(N - \begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} R \\ S \end{pmatrix} \right) \delta\Lambda = H - \begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} F \\ G \end{pmatrix}$$

Matrix form of the system

$$\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\tilde{\mathcal{N}}(\mathbf{x}_k)$$

By eliminating δU and δQ , one gets the **global** system

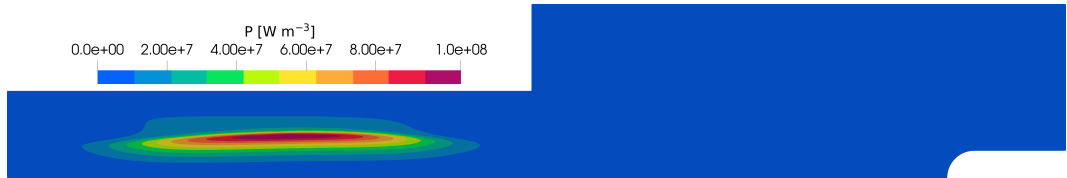
$$\left(N - \begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} R \\ S \end{pmatrix} \right) \delta\Lambda = H - \begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} F \\ G \end{pmatrix}$$

System solved for λ , then $\mathbf{u}$ and $\mathbf{q}$ are retrieved by small local matrices inversion.

$$\begin{pmatrix} \delta Q \\ \delta U \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} F \\ G \end{pmatrix} - \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} R \\ S \end{pmatrix} \delta\Lambda$$

The problem of power quenching

If the power dissipated is not maintained, the torch quenches!



$$P_{target} = \int_{NS} P_J dV$$

The problem of power quenching

If the power dissipated is not maintained, the torch quenches!

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v}) + \nabla p - \nabla \cdot \boldsymbol{\tau} = \mathbf{F}^L$$

$$\partial_t \left(\rho e + \frac{1}{2} \rho \|\mathbf{v}\|^2 \right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2} \rho \|\mathbf{v}\|^2 \mathbf{v} + p \mathbf{v} - \boldsymbol{\tau} \cdot \mathbf{v} - \mathbf{q} \right) = P_J$$

$$\Delta E_P = \frac{E_P}{r^2} + i2\pi f \mu_0 \sigma_e (E_C + E_P)$$

The problem of power quenching

If the power dissipated is not maintained, the torch quenches!

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v}) + \nabla p - \nabla \cdot \boldsymbol{\tau} = \gamma (\mathbf{F}^L)'$$

$$\partial_t \left(\rho e + \frac{1}{2} \rho \|\mathbf{v}\|^2 \right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2} \rho \|\mathbf{v}\|^2 \mathbf{v} + p \mathbf{v} - \boldsymbol{\tau} \cdot \mathbf{v} - \mathbf{q} \right) = \gamma P_J'$$

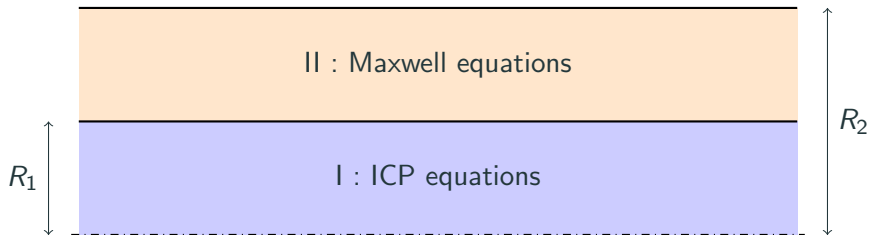
$$\Delta E_P' = \frac{E_P'}{r^2} + i2\pi f \mu_0 \sigma_e (E_C' + E_P')$$

with

$$E_P = \sqrt{\gamma} E_P' \quad E_C = \sqrt{\gamma} E_C', \quad \gamma = \frac{P_{\text{target}}}{\int_{NS} P_J' dV}$$

Results

Manufactured solution



$$p = \Delta p_0 f(z, r) + p_0 \quad T = T_{min} + (T_{max} - T_{min}) f(z, r)$$

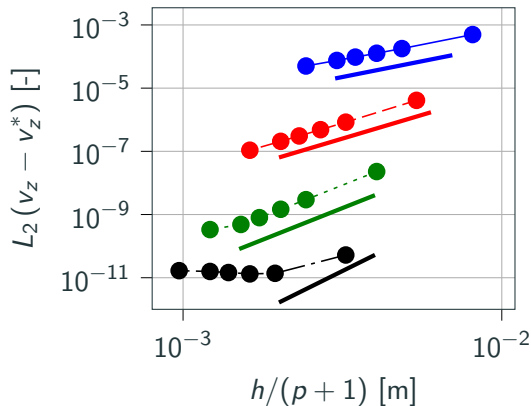
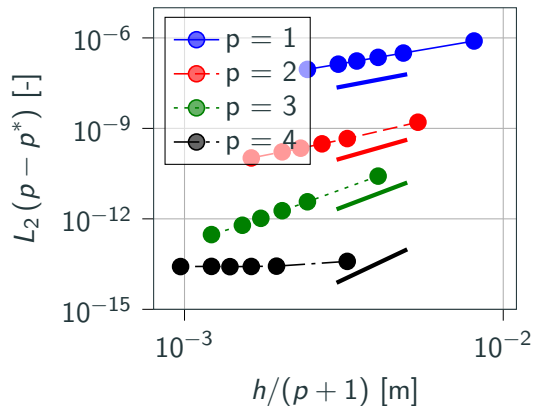
$$v_z = u_0 f(z, r) \quad E_p = E_0 f(z, r)(1 - i)$$

$$v_r = u_0 f(z, r) \quad v_\theta = u_0 f(z, r)$$

with

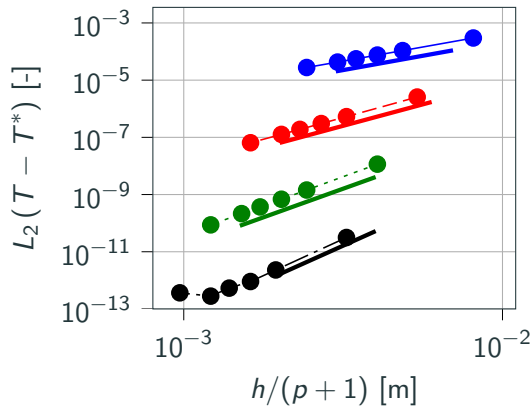
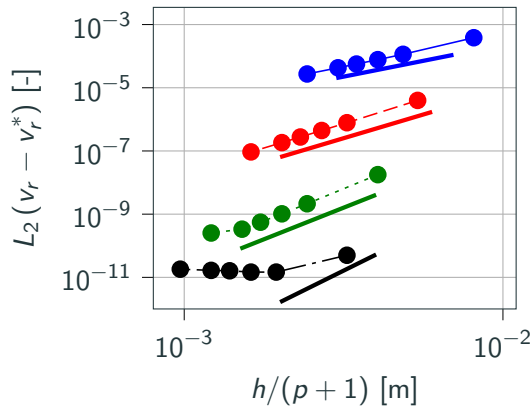
$$f(z, r) = \left(\frac{rz}{RL} \right)^2 \exp \left(-\frac{z}{L} - \frac{r}{R_1} \right)$$

Convergence study : in the torch



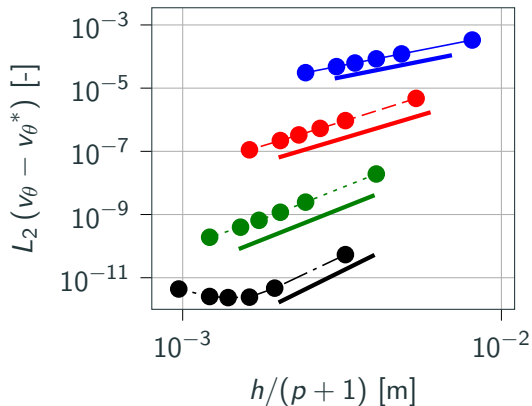
Convergence order of $p + 1$ is retrieved!

Convergence study : in the torch



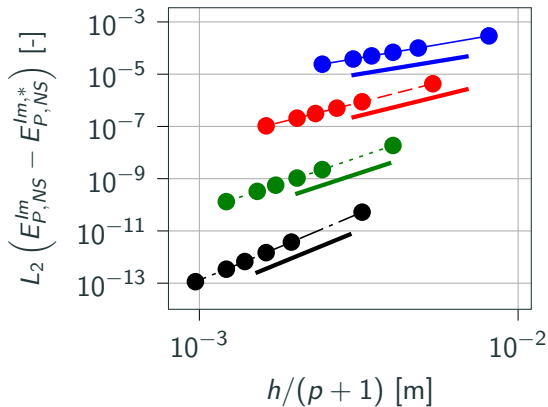
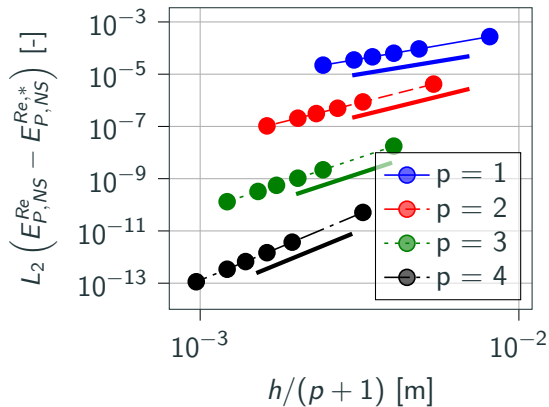
Convergence order of $p + 1$ is retrieved!

Convergence study : in the torch



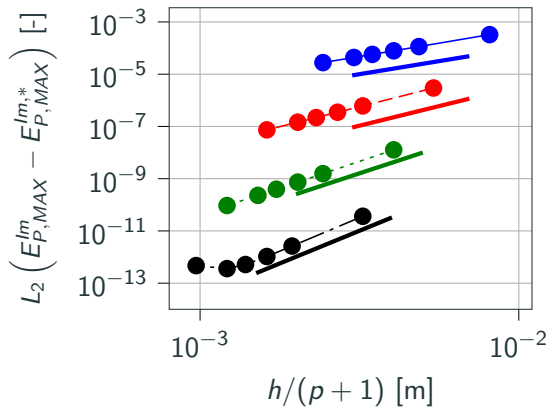
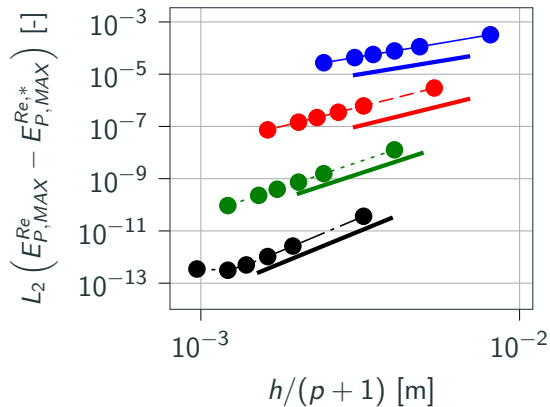
Convergence order of $p + 1$ is retrieved!

Convergence study : outside the torch



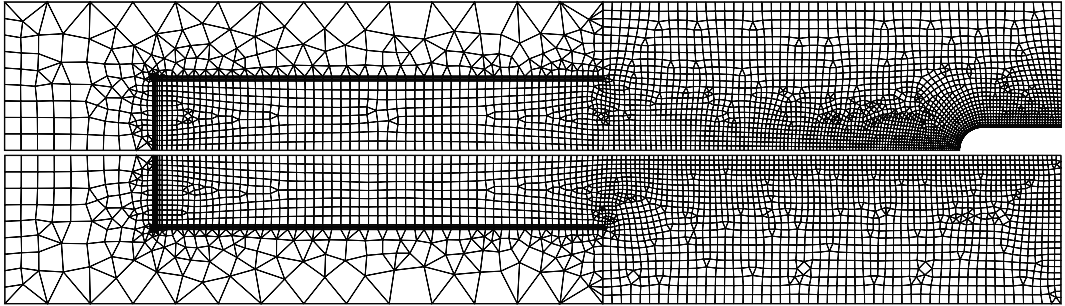
Convergence order of $p+1$ is retrieved!

Convergence study : outside the torch



Convergence order of $p+1$ is retrieved!

Mesh and order dependence study



This mesh at $p = 4$ with 4013 elements for the probe case and 3055 for the freestream case is sufficient for capturing the solution accurately.

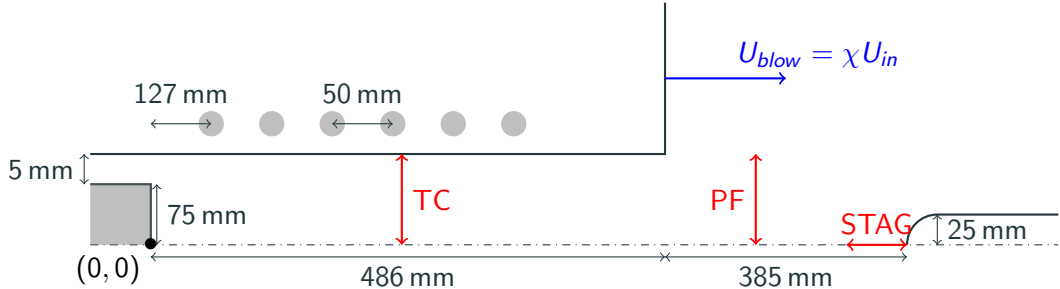
The mesh size close to the probe is $10\text{ }\mu\text{m}$ thick, while it is of $1\text{ }\mu\text{m}$ for FV!

The question of the coflow

A coflow is introduced in the chamber for obtaining a steady state, does it have an important impact?

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The question of the coflow

A coflow is introduced in the chamber for obtaining a steady state, does it have an important impact?

- $P = 100 \text{ kW}$
- $p_0 = 10\,000 \text{ Pa}$
- $f = 0.37 \text{ MHz}$
- $Q = 16 \text{ g s}^{-1}$
- $S = 20^\circ$
- air mixture with 11 species

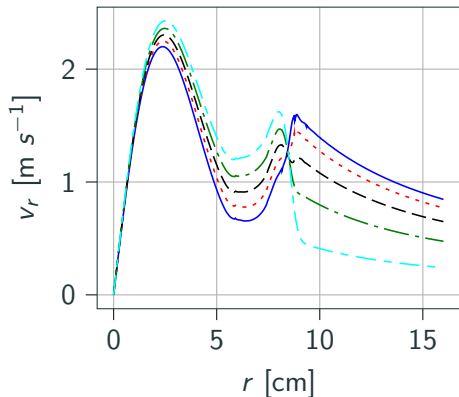
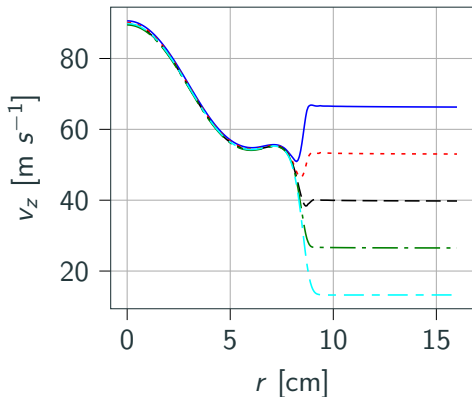
The coflow χ is defined as

$$\chi = \frac{U_{blow}}{U_{in}}$$

and tested from 0.2 to 1.0.

The question of the coflow

A coflow is introduced in the chamber for obtaining a steady state, does it have an important impact?



The question of the coflow

A coflow is introduced in the chamber for obtaining a steady state, does it have an important impact? Short answer is no, BUT...

... we were not able to obtain any steady state without coflow.

A question to ask is to see if steady state is possible in ICP?

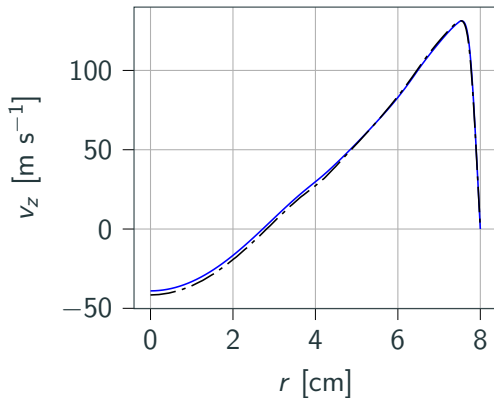
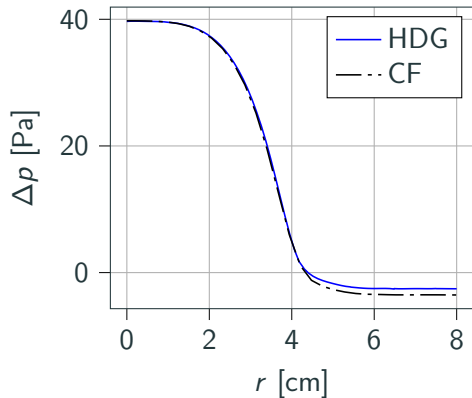
Left for future work...

We compared the results of the COOLFluid solver and the HDG code.

- $p_0 = 5000 \text{ Pa}$
- $Q = 16 \text{ g s}^{-1}$
- $T_{wall} = T_{inlet} = 350 \text{ K}$
- $P = 100 \text{ kW}$
- $f = 0.37 \text{ MHz}$
- air mixture with 11 species.
- No swirl angle.
- The coflow applied for HDG is $\chi_{HDG} = 0.2$.

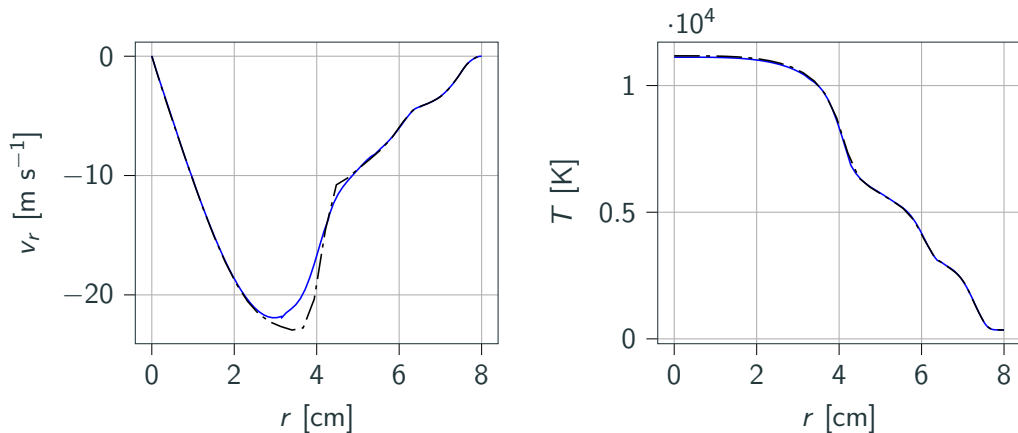
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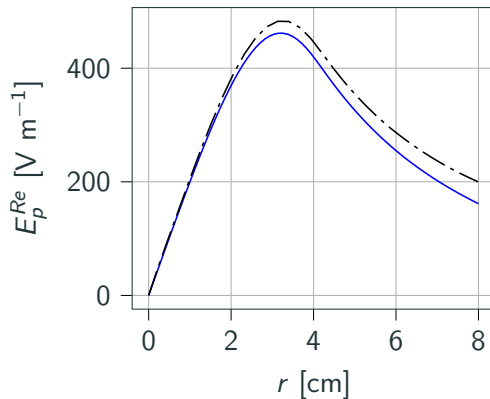
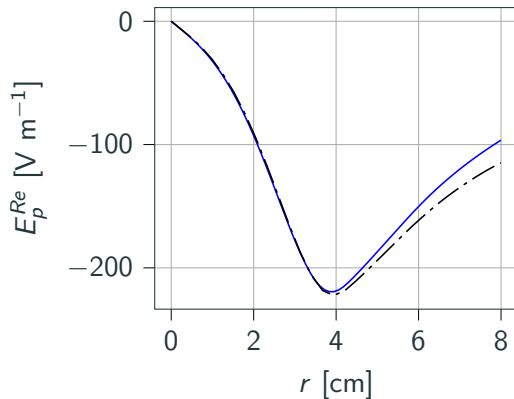
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Comparison with the COOLFluid solver

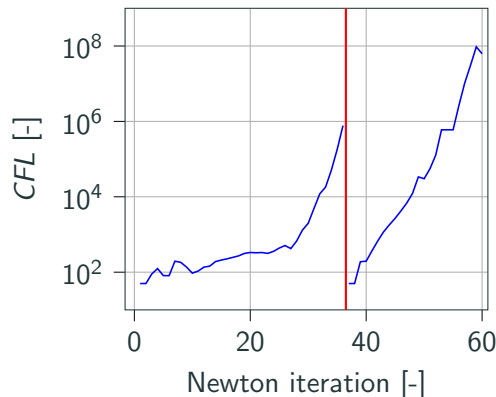
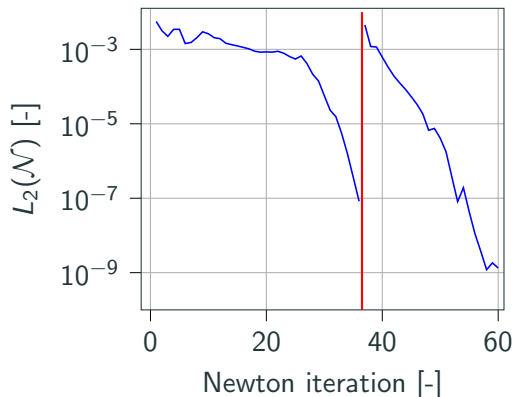
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Convergence history

We use **GMRES** solver with at most 50 vectors and a **ILU(4)** preconditioner.

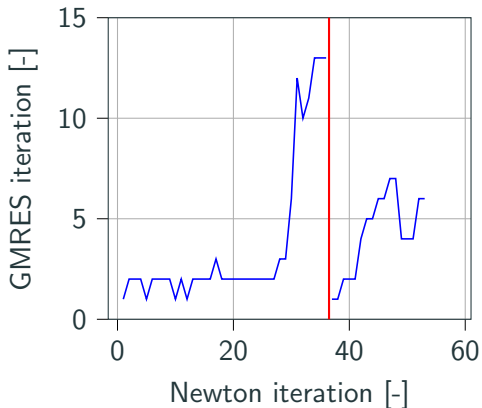
We first solve with polynomial degree 1 then 4 (red separation line).



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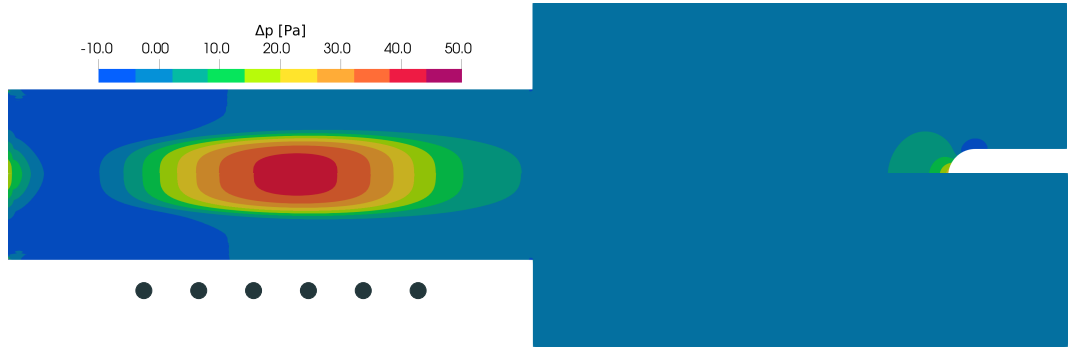
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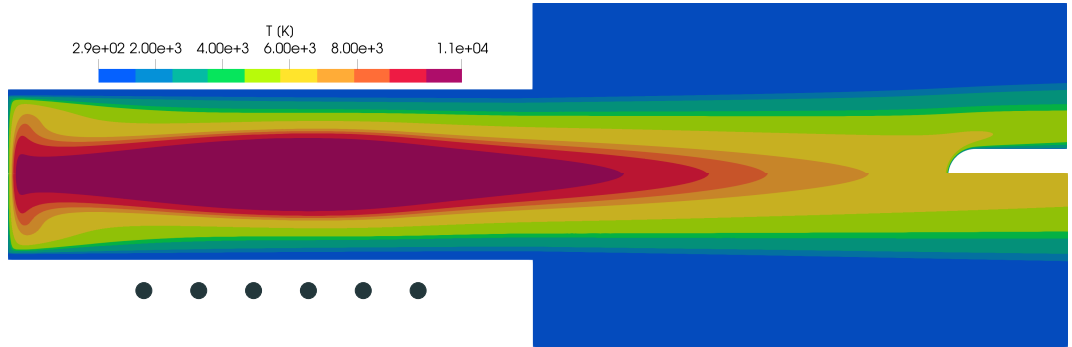
We have a fast and robust solver for ICP, able to produce meaningful results in 20 minutes of time on a single laptop using 8 OMP threads!

Flow fields in ICP: gauge pressure



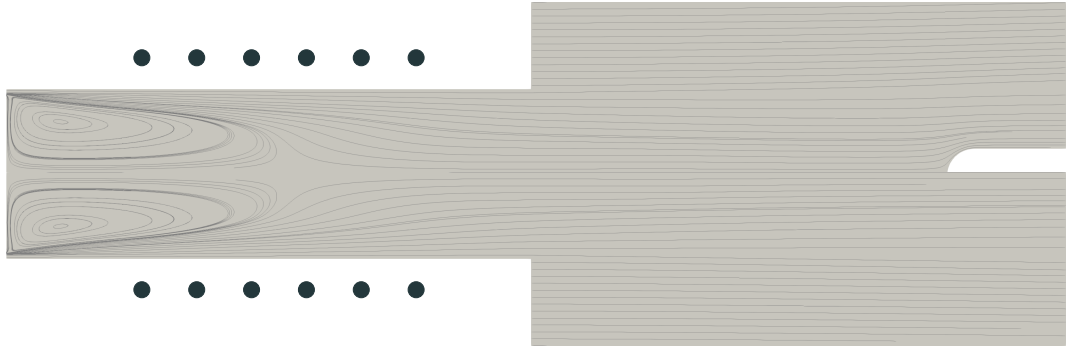
The pressure is almost constant everywhere ($p_0 = 10\,000\text{ Pa}$).

Flow fields in ICP: temperature



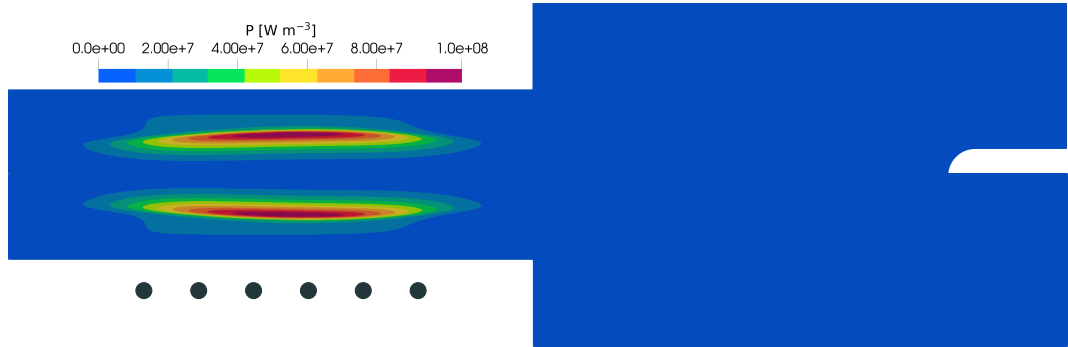
Large temperature gradients.

Flow fields in ICP: streamlines



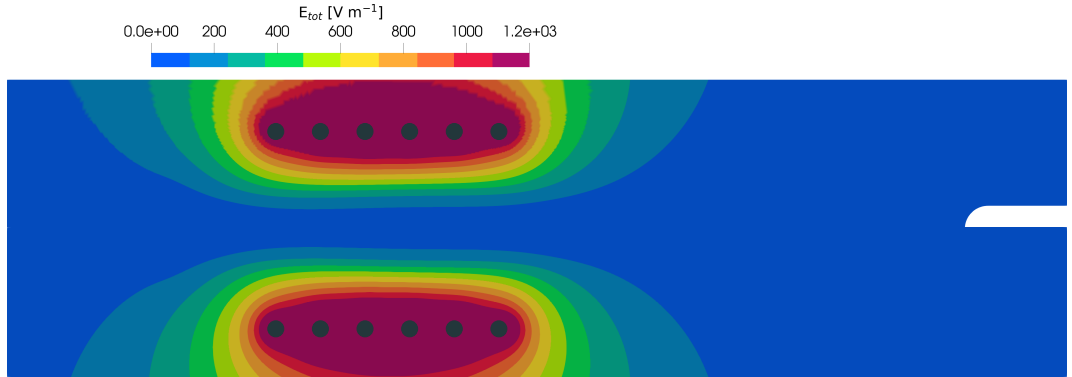
$$\nabla \cdot \mathbf{v} \simeq 0$$

Flow fields in ICP: volume power

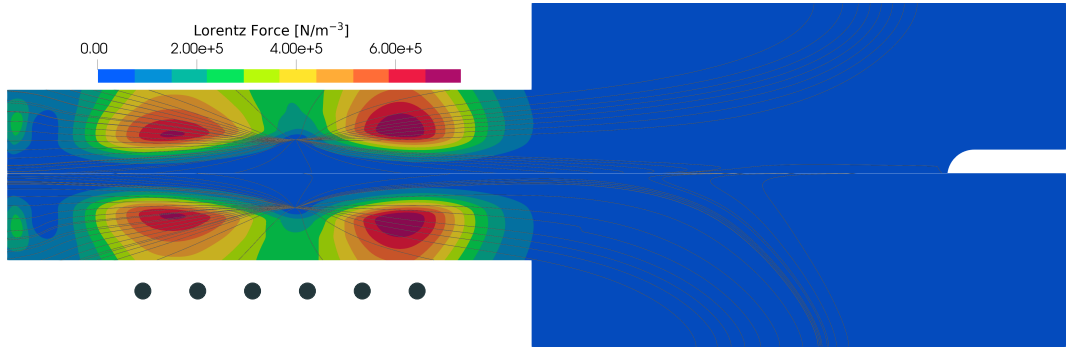


Volume power peak off-axis, where the coupling is the strongest.

Flow fields in ICP: Total electric field



Flow fields in ICP: Lorentz force



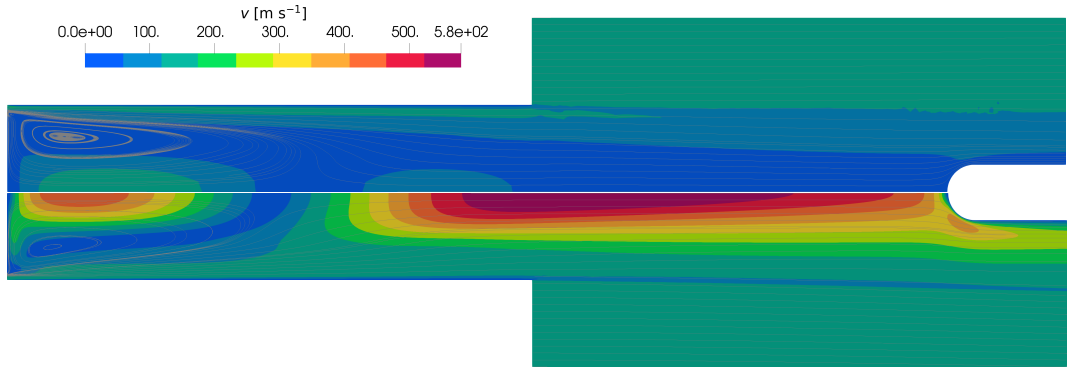
Parametric study

We performed a parametric study of ICP for the parameters used by the experimenters.

- Q from 8 g s^{-1} to 16 g s^{-1} .
- p_0 from 5000 Pa to 20 000 Pa.
- P_{tot} from 50 kW to 200 kW.
- f from 10 kHz, to 1 MHz.
- S from 0° to 20° .

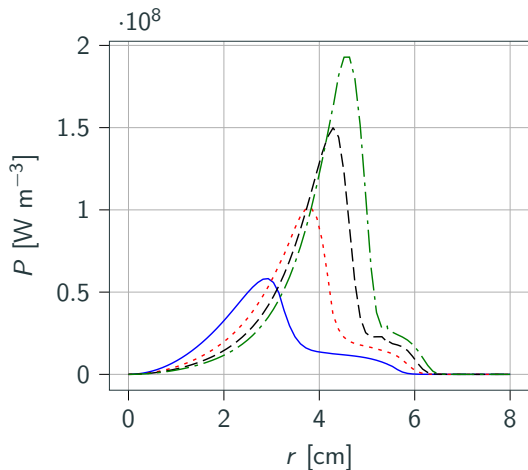
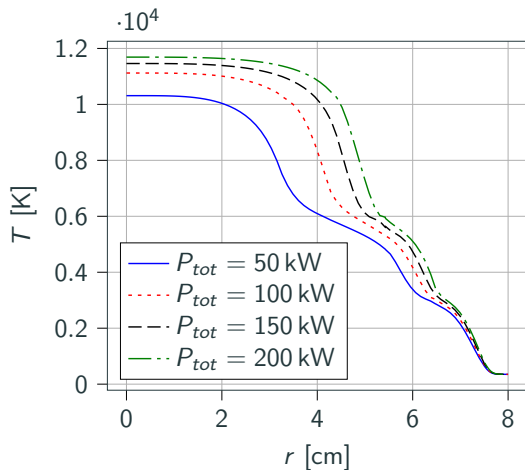
The complete discussion is in the thesis.

Power dissipated in the facility: impact on mass flow rate

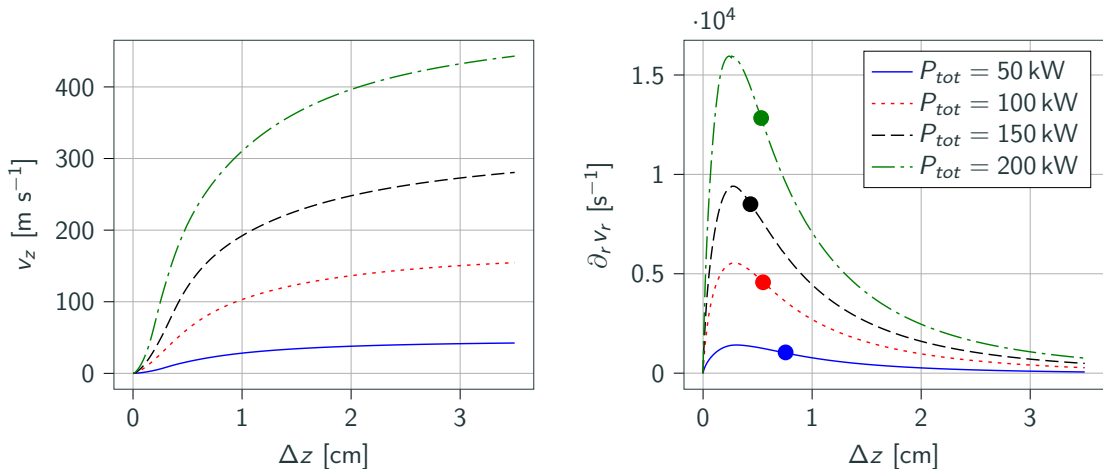


$P_{tot} = 50 \text{ kW}$ (top) and $P_{tot} = 200 \text{ kW}$ (bottom)

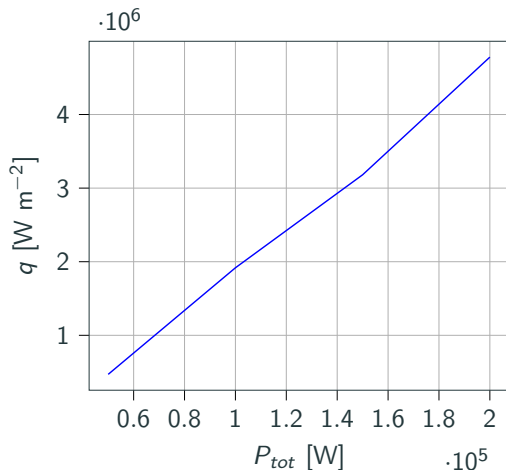
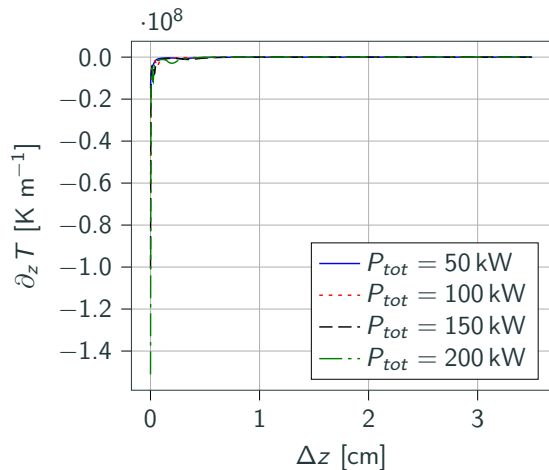
Power dissipated in the facility: impact in the torch



Power dissipated in the facility: impact along the stagnation line



Power dissipated in the facility: impact along the stagnation line



Conclusions and future work

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We believe so, but there is still much work to do...

Going 2D, 3D, and complex chemistry and radiation for unsteady simulations will need much more computational power.

For supersonic, shock capturing and redesign of the numerical flux.

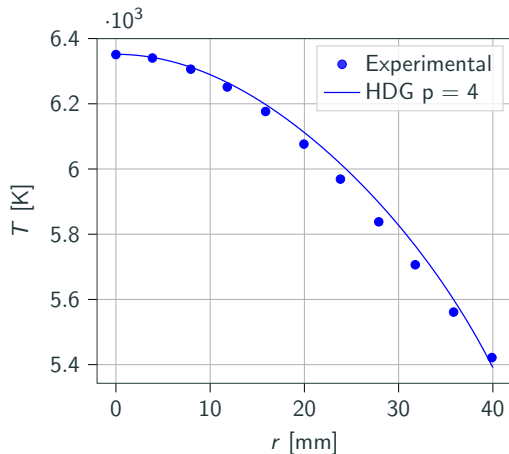
Future work

Main effort is to put our developments in the ARGO DG solver, which is fully paralleled and equipped shock capturing technologies. (Landry Riou)

We see the following roadmap:

- Going 3D.
- Going unsteady (also relieving the average over the induction current period).
- Development of chemistry (non equilibrium).
- Development of radiation.
- For all of the above, comparison with experiment when possible.

Comparison with experiment



- $p_0 = 200 \text{ mbar}$
- $Q = 16 \text{ g s}^{-1}$
- $P_{exp.} = 200 \text{ kW}$
- $P_{num.} = 78 \text{ kW}$
- Freestream, no swirl, air.

Comparison of the experimental $T(r)$ at 385 mm from the torch exit, from $r = 0$ to 40 mm.